Sri Harsha Mudumba

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SUMMARY

Machine Learning Engineer with 3+ years of experience building and optimizing production ML systems, with a focus on efficient language models and scalable inference. Experienced in transformer-based architectures, model evaluation, and performance-aware deployment. Combines systems engineering discipline with applied research to deliver reliable, low-latency AI solutions in real-world environments.

TECHNICAL SKILLS

Programming & Data Processing: Python, C++ (modern standards), SQL, Bash, Data Structures and Algorithms, Linux Systems.
ML Inference & Optimization: Inference optimization, latency and throughput profiling, memory efficiency, early-exit architectures, quantization (post-training and inference-aware), sparsity-aware execution, model benchmarking and performance validation.
Machine Learning Frameworks: PyTorch, TensorFlow, Hugging Face, ONNX Runtime, IREE.
Natural Language Processing & GenAI: Text Preprocessing, Tokenization, Named Entity Recognition (NER), Topic Modeling, Sentiment Analysis, Transformers, Hugging Face, LangChain Retrieval-Augmented Generation (RAG), Vector Databases (FAISS, Pinecone)
Deployment & ML Systems: FastAPI, REST-based model serving, batch and real-time inference pipelines, experiment tracking (MLflow), model versioning, CI/CD for ML systems, Docker, Kubernetes
Cloud & Data Engineering: AWS (S3, EC2, Lambda), Azure Machine Learning, ETL Pipelines, Batch & Real-Time Inference Optimization.

PROFESSIONAL EXPERIENCE

AI Engineer Intern, *HCLTech* *Jul 2025 – Present | Remote, USA*

- Contributed to end-to-end ML workflows from problem framing to feature engineering for internal and client AI solutions.
- Built transformer-based NLP and embedding pipelines, improving extraction accuracy by 20%.
- Prototyped semantic search and reranking systems, increasing top-k relevance by 25%.
- Implemented experiment tracking and model versioning using MLflow for reproducible training.
- Deployed batch and real-time inference services with FastAPI and Docker, adding latency monitoring and reliability logging.
- Partnered with data engineers and stakeholders to deliver measurable, production-ready ML features.

Software Engineer, *Cognizant Technology Solutions* *Aug 2020 – Jul 2023 |KAR, INDIA*

- Managed and optimized production databases (5+ TB scale) serving ML training and inference pipelines, maintaining 99.9% uptime under strict SLAs across 3+ years.
- Reduced query execution latency by ~30% through systematic profiling of execution plans, memory utilization, and I/O bottlenecks, directly improving throughput for compute-intensive ML workloads.
- Designed and maintained data ingestion workflows processing millions of records daily, ensuring correctness and timely availability for downstream ML feature pipelines.
- Collaborated with ML and infrastructure teams across 2+ projects to resolve data quality issues impacting model training runs, reducing pipeline failures and improving data reliability.

RESEARCH EXPERIENCE

DURACIM – RL-Guided Early-Exit Co-Design for Compute-in-Memory Systems | Reinforcement Learning, Python, PyTorch

- Designed sequential decision policies using PPO and Q learning to optimize accuracy latency and energy tradeoffs under deployment constraints.
- Built an experiment pipeline that integrates PyTorch model execution with CIMLOOP based cost modeling to estimate per layer energy and device endurance trends.
- Achieved **86 to 88 percent relative inference energy savings** on ResNet 50 compared to full depth execution while keeping accuracy within about one percentage point.
- Evaluated endurance behavior under multiple mapping configurations and policy choices, showing the effective RRAM lifetime **to 1.02 × 10^7 years** under the endurance model, even in all-RRAM configurations.

FEATURED PROJECTS

TorchWeave-LLM – Continuous Batching Inference Server | Python, PyTorch, FastAPI, Docker **(Link)**

- Developed a continuous batching inference server for LLMs using asynchronous scheduling, achieving 4x throughput improvement and 35% lower latency under concurrent workloads.
- Designed a scalable request orchestration layer to efficiently manage dynamic batching, improving GPU utilization and stabilizing tail latency under load.

LEXA – Lightweight Local RAG with TinyLlama 1.1B | CUDA, PyTorch, FastAPI, Sentence Transformers **(Link)**

- Built a GPU-optimized local RAG system integrating TinyLlama with adaptive early-exit logic, reducing average inference latency by 40% and GPU power consumption by 30%.
- Implemented efficient retrieval, caching, and memory-aware execution strategies to enable stable low-latency performance on resource-constrained systems.

EDUCATION

Master of Science, *Iowa State University* Aug 2023 – Aug 2025 | IA, USA
Computer Engineering (Computing and Networking Systems)

Bachelor of Technology, Amrita Vishwa Vidyapeetham Aug 2016 – Aug 2020 | INDIA
Electronics and Communication Engineering